N225 Statarb

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Industry Classification vs. k-Means Pair Discovery
Proposed LLM News-Screening Layer

ABSTRACT

Statistical arbitrage on equity pairs rests on 1 assumption: that 2 historically correlated stocks will revert to a common spread after a temporary divergence. This paper documents an empirical study of pair selection on the Nikkei 225 universe, comparing 2 approaches under identical signals, position sizing, and transaction costs: classical industry classification, and unsupervised k-Means clustering on return correlations. Industry classification produces a portfolio total return of +1.24% (Sharpe 0.083, 3/5 pairs winning); k-Means produces –0.42% (Sharpe 0.002, 1/5 pairs winning).

The study surfaces a failure mode that cuts across both strategies. The 1 overlapping pair — Sojitz (2768.T) and Sumitomo (8053.T), both sogo shosha — scored highest on cointegration strength (ADF: –3.90, p<0.01) and correlation (0.969), yet produced the worst back-test result once a copper-mine acquisition by Sumitomo structurally decoupled the spread. This failure is selection-method-agnostic: it cannot be diagnosed from price data because it originates in a corporate news event.

The 2nd contribution is a proposed experimental design for an LLM news-screening layer whose sole job is to flag pair-breakdown events before they destroy capital. The claim is narrow and testable: does interposing an LLM news filter between the z-score signal and position sizing reduce drawdown during structurally-driven spread divergences, relative to a pure-quant baseline?

1. Introduction

A pair trade produces a signal — an entry z-score — and the system must decide whether to act on it. In most implementations the answer is mechanical: if |z| > threshold, enter; if |z| < exit-threshold, close. This works when spread divergences are mean-reverting noise. It fails when they are structural breaks in disguise.

A structural break looks, in the 1st days, exactly like a normal widening. Price data does not separate the cases. What separates them — often — is a news event: an acquisition, a regulatory change, a commodity price shock specific to 1 leg. This information exists in text, not in the cointegration statistic — and is not solved by choosing a better pair-selection method. Both industry classification and clustering will select a pair if the statistical signal is strong enough.

LLMs can read financial news and judge whether a specific corporate event is likely to affect the long-run relationship between 2 companies. The question this paper proposes to test: is that judgment, applied at entry time, profitable enough to justify the added complexity?

2. Data and Universe

Universe	188 N225 constituents, avg. daily volume > 1 M shares
Period	January 2023 – May 2025 · 589 trading days
Source	Yahoo Finance (yfinance)
Lookback	90-day rolling window · min. 60 common days (holiday-adjusted)

3. Pair Selection

Strategy 1 — Industry Classification

Candidate pairs are formed within industry groups. Of 63 N225 industry groups, 23 contain only 1 constituent and are excluded. All within-industry combinations of the remaining groups are scored and ranked. Rationale: shared macro drivers, supply-chain exposure, and regulatory environment provide a fundamental basis for cointegration. Limitation: cross-sector pairs with strong statistical cointegration are structurally excluded.

Strategy 2 — k-Means Clustering

Input	90-day rolling daily return matrix (stocks × days)
Algorithm	k-Means++ with elbow method · optimal k = 8 (N225)
Pairs from	Intra-cluster combinations only

Clustering groups stocks by geometric similarity of return trajectories, regardless of industry. Of the top-5 pairs selected by each strategy, 1 pair overlapped: Sojitz (2768.T) vs. Sumitomo (8053.T).

Scoring Formula and Trading Parameters

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Score = ADF weight + 2.0 × |Pearson r|   [ ADF weight: 3 (1%) · 2 (5%) · 1 (10%) ·
0 (none) ]   Top-5 pairs selected.
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Spread · Z-score window	$Price_1 - Price_2$ · 30-day rolling
Entry / Exit thresholds	$\lvert z\rvert > 1.5$ entry · $\lvert z\rvert < 0.4$ exit · 1-day signal lag
Capital · Costs	JPY 1,000,000 per pair · 0.15% per leg (entry + exit) · 50/50 long-short

4. Results

Metric	Strategy 1 — Industry	Strategy 2 — k-Means
Portfolio total return	+1.24%	–0.42%
Sharpe ratio	0.083	0.002
Pairs winning	3 of 5	1 of 5
Pair overlap	—	1 of 5

Industry classification outperforms k-Means by 1.66 pp on the same signal, same capital, and same transaction-cost assumption. Within-industry pairs share a denser set of common fundamental drivers — regulation, supply chains, customer bases — that support tighter, more durable spreads. The cross-sector pairs discovered by k-Means co-moved statistically in the formation window but proved less robust in the back-test, with only 1 of 5 pairs profitable. The result does not settle the general question — the back-test is short and the sample small — but it establishes that statistical co-movement alone is not a sufficient basis for durable cointegration.

5. The Sojitz / Sumitomo Failure

Pair	Sojitz (2768.T) / Sumitomo (8053.T) — both sogo shosha
ADF / Pearson r	–3.90 ($p < 0.01$) · $r = 0.969$
Score	Highest of all pairs under both strategies
Outcome	Worst back-test result in the portfolio under both strategies

Both companies are diversified Japanese trading houses with overlapping commodity, energy, and industrial businesses. The statistical relationship was extraordinarily strong in the formation window. During the back-test period, Sumitomo announced a material copper-mine acquisition — shifting its commodity exposure, changing its capital allocation profile, and permanently altering the long-run relationship between the 2 companies' earnings streams. The spread did not revert.

> **Key insight:** The information that this divergence was structural was in news text at the time of signal generation — not in prices. This failure is selection-method-agnostic: industry classification did not protect against it; k-Means did not protect against it. Neither can, because neither reads news. This is the gap the proposed LLM layer addresses.

6. Proposed Architecture: LLM News Screening

The LLM layer intercepts the signal pipeline at 1 decision point: between z-score threshold breach and position entry. It answers 1 question:

> Does recent news about either leg of this pair suggest the spread divergence may be structural rather than mean-reverting?

Step	Component	Description
1	**Z-Score Signal**	$\|z\| > 1.5$ threshold breached
2	**LLM News Filter**	Query prior 7 days of news for each ticker
3	**PASS / FLAG**	Structural break detected?
4	**Position Entry**	Enter if PASS · suppress if FLAG

LLM inputs: ticker identifiers, news headlines + summaries (prior 7 days), 1-line description of the assumed fundamental relationship. Output: PASS or FLAG, plus an optional 1-sentence reasoning trace.

What Gets Flagged

FLAG — Potential Structural Break	PASS — Expected to Mean-Revert
M&A / acquisition; divestiture / spin-off	Earnings beat/miss; analyst upgrade/downgrade
Regulatory action specific to 1 leg	Dividend announcement
Commodity exposure shift; credit event	Macro events affecting both legs equally

Implementation: financial newswire API (headlines + summaries sufficient); instruction-following LLM with financial domain competence; 1 query per entry signal (O(hundreds of tokens) — rounding error vs. trading costs).

7. The Proposed Experiment

Null hypothesis: adding an LLM news-screening layer produces no statistically significant improvement in risk-adjusted return over the pure-quant baseline.

Strategy 1+ (Baseline)	Industry classification + z-score signal · no news filter · stronger performer on the back-test period
Strategy 1-L (Treatment)	Identical pair selection and signal · LLM screening at entry · flagged signals suppressed
Validation	Walk-forward on a held-out period not used in pair-formation or scoring calibration

Metric	Why It Matters
Net return (1-L vs. 1+)	Does screening add alpha?
Max drawdown (1-L vs. 1+)	Does screening reduce tail events?
Win rate on flagged pairs	Were flagged pairs actually structural breaks?
False-positive rate	How often did screening suppress profitable entries?
Sharpe ratio (1-L vs. 1+)	Risk-adjusted comparison

> **Honest baseline:** if Strategy 1-L does not outperform Strategy 1+ on the held-out period net of costs, the LLM layer should be removed. A reasonable success target: the layer prevents 2–3 Sojitz/Sumitomo-style events per year per 5-pair portfolio, without suppressing more than 10–15% of profitable entries.

8. Discussion

Most proposed LLM applications in finance fall into 1 of 2 failure modes:

- The forecasting mistake: using LLMs to predict price direction — no principled advantage over quantitative methods.
- The integration mistake: proposing LLMs as a general orchestration layer with no clear, testable claim.

This proposal avoids both. The task — classifying whether a news event is a structural break — gives LLMs a genuine comparative advantage. Price data cannot read a news article; an LLM can. The classification boundary is well-defined enough to evaluate against ground truth.

The LLM layer is pair-selection-agnostic. Whether pairs are chosen by industry grouping or k-Means, any pair that triggers an entry signal is a candidate for screening. The narrow claim: the LLM helps at the 1 moment when the signal fires, by reading news that the quantitative model cannot read.

9. Limitations

Back-test is small	1 589-day period on 1 equity universe. A 1.66 pp differential is directional, not deployment-grade.
LLM layer not tested	Section 6 is a proposed design. Claims about LLM effectiveness are conditional on empirical validation.
News coverage uneven	Large-cap N225 names have adequate coverage. Smaller names or non-Japanese markets may suffer sparsity.

Language / translation	Japanese-language news may require a multilingual LLM or translation step — additional cost and error risk.
LLM hallucination	Model may flag non-events or miss real events. False-positive and false-negative rates must be measured.

10. Conclusion

We adopted k-Means clustering to move past simple industry classification toward a more sophisticated, data-driven method of pair discovery — 1 that groups stocks by the geometry of their return trajectories rather than by sector labels. On the N225 universe over the study period, that bet did not pay off: industry classification outperformed k-Means by 1.66 pp, consistent with the hypothesis that shared fundamental drivers, not statistical co-movement alone, underpin durable spread relationships. The more sophisticated selection method did not beat the simpler, fundamentals-based one.

Neither strategy protects against the failure mode identified here: a structurally-driven spread divergence caused by an idiosyncratic corporate news event. The Sojitz/Sumitomo pair — the highest-scoring pair under both strategies — failed for exactly this reason. The information needed to avoid the loss was in the news, not in the prices.

The proposed LLM news-screening layer addresses this gap at 1 decision point, with a narrow and testable claim. If the LLM layer does not improve risk-adjusted return on the held-out period, it should not be deployed.

References

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